DONALD AMADI

Senior Product Engineer | Full Stack: React, Node.js/TypeScript, Mobile | Fintech | AI-Forward

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Professional Summary

Senior Product Engineer with 5+ years in fintech, shipping products across **web, backend and mobile**. I built **PinDey** (live at pindey.app) end to end, in one TypeScript monorepo: the **React/Next.js** public site and browser tools, a **React/Next.js** admin back office, and the **Node.js/TypeScript** services behind them (REST API, WebSocket gateway, BullMQ worker on **PostgreSQL** and Redis, with a **transactional outbox** and **idempotency keys**), plus the cross-platform mobile clients. In fintech I build payment and lending flows around the failure states that matter in money movement: retries, reconciliation and idempotent disbursement. I bring 5 years of production cross-platform mobile work to **React Native**, and I ship AI-forward through agent loops and skill files I author.

Core Skills

Frontend: React, Next.js (App Router, server components), TypeScript, Tailwind CSS, accessible UI (axe-core), Playwright end-to-end tests, Vitest, Cloudflare (OpenNext) deployment.
Mobile: cross-platform apps in production for 5+ years (Flutter, Android, iOS widgets in Swift/SwiftUI); React Native through the same React + TypeScript model I use on the web; push, deep links, biometrics, offline-tolerant networking.
Backend: Node.js, TypeScript, NestJS, Fastify, Express, WebSockets, RESTful API design, PostgreSQL + PostGIS, SQL and versioned migrations, Prisma, Redis, BullMQ job queues, Supabase, JWT with refresh-token rotation, OAuth (Apple, Google), email OTP, push fan-out.
Architecture: event-driven systems with a transactional outbox, idempotency keys, ledger-safe retries and reconciliation, state machines, RBAC and audit logging, data retention by design, Clean Architecture.
Payments & fintech: wallets and ledgers, disbursement and card repayment (Rave/Flutterwave, Paystack), open banking (Mono), KYC (Prembly, HyperVerge; BVN/NIN/liveness), AES request/response encryption.
Testing & ops: automated tests, GitHub Actions, Sentry, GCP, Vercel, Supabase.
Languages: TypeScript, JavaScript, SQL, Dart, Swift, Java, Python.

AI-Forward Engineering

How I ship: agentic loops (plan, act, verify, correct) with explicit guardrails and stop conditions. Coding agents implement, refactor and review; static analysis, tests and my own review gate every merge.
Harness & context engineering: skill files and CLAUDE.md/AGENTS.md hierarchies that let agents work inside real conventions; decomposed a live fintech codebase into feature-scoped skill documents and published the approach as an open-source CLI (flutter_skill_gen). Tools: Claude Code, Cursor, Copilot, MCP servers, Claude API.

Education

University of Lagos **August 2023**
Bachelor of Science in Computer Engineering *Lagos, Nigeria*

Experience

PinDey | pindey.app **September 2026 – Present**
Founder & Full-Stack Engineer, own product (Web, Back Office, API, Mobile) *Lagos, Nigeria*

- Designed, built and run **PinDey**, a live friend-finding product, full stack in a pnpm/Turborepo TypeScript monorepo: **React/Next.js** web apps, **Node.js** services (NestJS on Fastify) on **PostgreSQL + PostGIS** and Redis, and the iOS/Android clients, all built against one shared OpenAPI contract.
- Built the public site at pindey.app in **React and Next.js** (App Router) with Tailwind, deployed to Cloudflare via OpenNext: store-badge landing, an interactive walkthrough, invite and event deep links that survive the app install, legal pages, and no-install browser tools (drop a meet point, a crowd beacon) whose location lives only in the URL fragment and never reaches the server.
- Built the admin **back office in React/Next.js** with server-only data access over the `/admin` API: users, Spaces, reports and moderation, testers, remote config, analytics, admin roles and an audit log, with authenticator-app two-factor sign-in.
- Ran real-time presence through a **Node.js WebSocket gateway** and a **BullMQ** worker for push fan-out, with Playwright end-to-end, axe-core accessibility and Testcontainers-backed integration tests gating changes across web and API.
- Made real-time delivery reliable with a **transactional outbox**: state changes and their publish events commit in one transaction, and a relay publishes with attempt counts and last-error tracking, so a crash between write and publish can never drop or invent an update.

- Protected every mutating endpoint with **idempotency keys** (request hash + stored response) so mobile retries over flaky networks replay the original result instead of double-applying; sessions use **rotating refresh tokens**, federated identities (Apple, Google) and email OTP.

Zedcrest Group **March 2024 – Current**

Mobile Engineer, Full-time (Zedvance Personal & Business Banking) *Lagos, Nigeria*

- Built **ExtraCash**, embedded digital lending inside Zedvance: eligibility from open-banking bank-statement analysis (Mono), KYC tiers and employment/income capture, with automated decisioning and reason-carrying limit reassessment when income changes.
- Modelled the loan journey as a **resumable, persisted state machine** with capability gating: a user who drops at statement consent or KYC returns to the exact step, and deep links, push notifications or app resume can never skip a gate.
- Implemented the money flow end to end: instant disbursement to wallet, retried against Zedvance's wallet ledger with **idempotent keys**; repayment via debit card and Rave/Flutterwave **reconciled against the card rail** instead of optimistic local balances; OTP/email verification and KYC-upgrade gating.
- AI-forward delivery: turned 20 Figma screens for the ExtraCash revamp into a 1,852-line design spec and a single orchestrated agent brief covering 15 implementation areas, then built against it with agents under review.

SnapPay **April 2026 – June 2026**

Flutter Engineer, Contract (SnapPay) *Remote (Nigeria)*

- Delivered features for a production Nigerian fintech app (wallets, transfers, airtime/data/electricity bill payments, group & personal savings, invoices, QR pay, ticketing, rewards), shipping backwards-compatible, feature-flag-gated changes to live users.
- Engineered an AES-encrypted networking layer with request/response encryption and automatic encryption-key refresh-and-retry on HTTP 403, plus session-expiry auto-logout on HTTP 401.
- Integrated Prembly KYC (BVN/NIN/liveness), biometric authentication, QR scan-to-pay, PDF receipt generation and deep linking, on a layered MVVM core (Provider, get_it, dartz functional error handling).
- Authored modular, AI-loadable skill documents decomposing the codebase by feature so AI agents could work safely within it, applying harness and context engineering to a live app.

MedPal **June 2025 – Present**

Flutter Engineer, Contract (MedPal Provider) *Remote*

- Built real-time provider-to-patient video and voice consultations with the Agora RTC engine, wired into the full telemedicine session lifecycle (incoming request, accept, in-call controls, completion) for a healthcare-provider platform.
- Integrated Firebase AI (Vertex AI / Gemini) to power AI-assisted medical-specialty selection during provider onboarding.
- Delivered provider earnings and payouts (available balance), payments, availability scheduling, QR generation, biometric login, and document/consent capture.
- Led full English/Spanish internationalisation, migrating every user-facing string to gen-l10n (480+ keys) with CI parity tests, across a Clean Architecture, Cubit (flutter_bloc), go_router and get_it codebase with multi-flavor Firebase (dev/staging/prod).

Equivator **October 2024 – May 2025**

Senior Frontend Developer, Contract (Uplines Web/Mobile) *Saudi Arabia (Remote)*

- Contributed to web and mobile platforms for the Uplines project, including native iOS widgets (Swift/SwiftUI) that boosted daily active users by 10% and UX enhancements that increased engagement by 20%.
- Integrated advanced analytics (Firebase, CleverTap), enabling 30% faster data-driven decisions, and partnered with backend/design teams to improve scalability by 15%.
- Spearheaded new features (gift points, tutorials, onboarding) and resolved critical bugs via Sentry and Firebase Crashlytics, improving stability by 30%.

MAX **March 2024 – December 2024**

Flutter Mobile Developer, Full-time (PAMS Mobile, Falcon Web App) *Lagos, Nigeria*

- Re-architected a legacy codebase to Clean Architecture, cutting bugs by 60% and improving scalability.
- Built a multilingual testing app with voice playback, improving accessibility for diverse demographics.
- Built and enhanced Falcon web app's real-time map interface to track and monitor riders/drivers, integrating live geolocation, speed, and distance data for improved operational visibility.

Creative Advanced Technologies **June 2022 – August 2024**

Flutter Mobile Developer, Contract (Tamata, Taza) *Dubai, United Arab Emirates*

- Delivered a 25% UX improvement and 40% performance gain in an e-commerce app serving 500,000+ users through RxBloc optimisation and targeted performance tuning.

- Improved workflows by leading code reviews and task tracking (Bitbucket, Jira) and deploying a Bitrise CI/CD pipeline, cutting release times by 50% and achieving a 90% build success rate.
- Integrated Shorebird for code push, reducing deployment time by 70% and enabling faster delivery of updates on Android and iOS.

Platnova **October 2023 – April 2024**

Flutter Mobile Developer, Full-time (Platnova) *Lagos, Nigeria*

- Built a multi-currency wallet (5,000+ monthly transactions) with Paystack integration, improving efficiency and engagement.
- Led UI enhancements including a dark mode and design review contributions, boosting retention.

First Bank of Nigeria **February 2022 – January 2024**

Flutter Mobile Developer, Contract (LitApp) *Lagos, Nigeria*

- Built core flows of a multifunctional banking app: self-transfer, inter/intra-bank transfers, airtime purchases and bill payments, contributing to a 35% improvement in user experience and a 50% rise in customer satisfaction ratings.
- Iterated on usability with the UI/UX team, driving a 30% increase in session duration, a 15% improvement in retention and 10% more positive app reviews.

Student Paddy **June 2021 – June 2022**

Lead Mobile Developer (StudentPaddy) *Lagos, Nigeria*

- Led the build of an educational social app for iOS and Android in Flutter, with real-time in-app chat, Firebase push notifications and deep linking to drive engagement.

Algorism Limited **May 2021 – November 2021**

Software Developer Intern (RealtyProx, TaskProx) *Lagos, Nigeria*

- Designed a real-time bidding system for real estate and built reusable, performance-optimised Flutter components for property transactions.

Projects & Open Source

Job scraper platform | *React, Next.js, TypeScript, Node.js/Express, Prisma, PostgreSQL, Redis, BullMQ* **2026 – Current**

- Building a full-stack TypeScript job-aggregation platform: a **Next.js (App Router) and React** frontend with Tailwind for the dashboard, filters, CV and document pages, over a **Node.js/Express** REST API with Prisma on PostgreSQL, Redis and BullMQ queues for scheduled scraping jobs with retries, and JWT authentication on a typed API contract.

Adduce | *SaaS, PostgreSQL, API design* **August 2026 – Current**

- Founding an early-stage SaaS that produces signed, clause-mapped accessibility-compliance evidence (EN 301 549 / European Accessibility Act). Authored the architecture, data model, API, security model and decision log; Postgres provisioned in the EU.

Ofin Eko | *React Native, iOS and Android* **January 2025 – July 2025**

- Built and shipped a bilingual (English/Yorùbá) **React Native** app on my own for reading and listening to Lagos State law, live on the App Store and Google Play: offline-first caching so it works without a connection, text-to-speech for every statute, and deep links to any law or bookmark.

Leadership & Community

Melodia Academy, Lead Tutor (2022): mentored a team of 3 engineers in Flutter and ran code review on their projects.
Andela Hackathon for Justice (2021): built a prison record-keeping app to tackle pretrial overcrowding.
Writing: technical articles on Level Up Coding (Medium), most recently on context engineering for AI coding agents (2026).